

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

اللَّهُمَّ إِنِّي أَسْأَلُكَ عِلْمًا نَافِعًا،

وَرِزْقًا طَيِّبًا، وَعَمَلًا مُتَقَبَّلًا،

Multiplication Rule

Week 5

Independent Events

Unrelated

Recognize:

- Height greater than 6 feet and Weight more than 200 pounds
- Height greater than 6 feet and ownership of red car
- Weather condition and availability of route
- Chances to win a game and shoe size
- Marks obtained in quiz and eye color

Independent Events

Occurrence/ non-occurrence of one event have no effect on other

Independent Events

Two events A and B are said to be **independent** events if

$$P(A|B) = P(A), \quad P(B|A) = P(B), \quad \text{and} \quad P(A \cap B) = P(A) P(B)$$

Example: 1

A small town has one fire engine and one ambulance available for emergencies. The probability that the fire engine is available when needed is 0.98, and the probability that the ambulance is available when called is 0.92. In the event of an injury resulting from a burning building, find the probability that both the ambulance and the fire engine will be available, assuming they operate independently.

$P(\text{Ambulance and Fire Engine}) = P(A)P(F)$
$= (0.92)(0.98)$
$= 0.9016$

Example: 2

Consider a simple chain of two links as in fig. Let E_1 be the breakage of link 1 and E_2 be the breakage of link 2. If the fracture strength of a link is less than 300 kg, it will fail by fracture. Suppose that prob. Of this happening to either of the two links is 0.05. Find the probability of failure of the chain. Assume that E_1 and E_2 are statistically independent.



P(failure of chain)=

$$\begin{aligned} P(E_1 \cup E_2) &= P(E_1) + P(E_2) - P(E_1 \cap E_2) \\ &= 0.05 + 0.05 - P(E_1)P(E_2) \\ &= 0.10 - (0.05 * 0.05) \end{aligned}$$

Example: 3

A town has two fire engines operating independently. The probability that a specific engine is available when needed is 0.96.

- i. What is the probability that a fire engine is available when needed.
- ii. What is the probability that neither is available when needed.

P(Fire Engine is available)=

$P(FE_1 \cup FE_2) = P(FE_1) + P(FE_2) - P(FE_1 \cap FE_2)$
$= 0.96 + 0.96 - P(FE_1)P(FE_2)$
$= 1.92 - (0.96 * 0.96)$

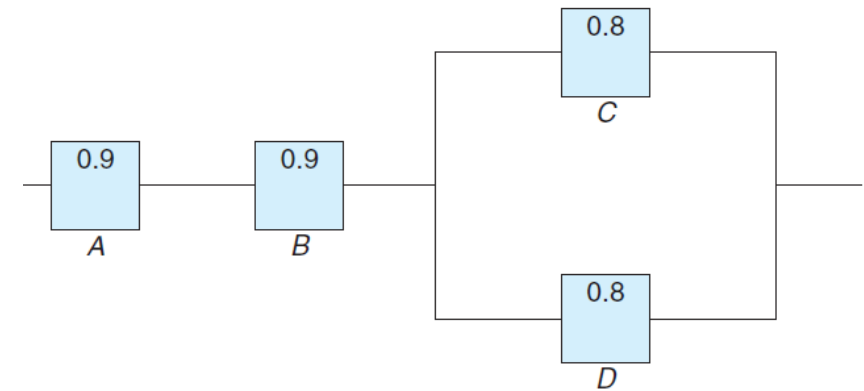
P(Neither is available)=

$P(FE'_1 \cap FE'_2) = P(FE_1 \cup FE_2)'$
$= 1 - P(FE_1 \cup FE_2)$

Example: 4

An electrical system consist of four components. The system works if components A and B work and either of the components C or D works. The reliability of each component is also known. Assume that the four components work independently. Find the probability that

- i. The component C does not work, given that the entire system works.

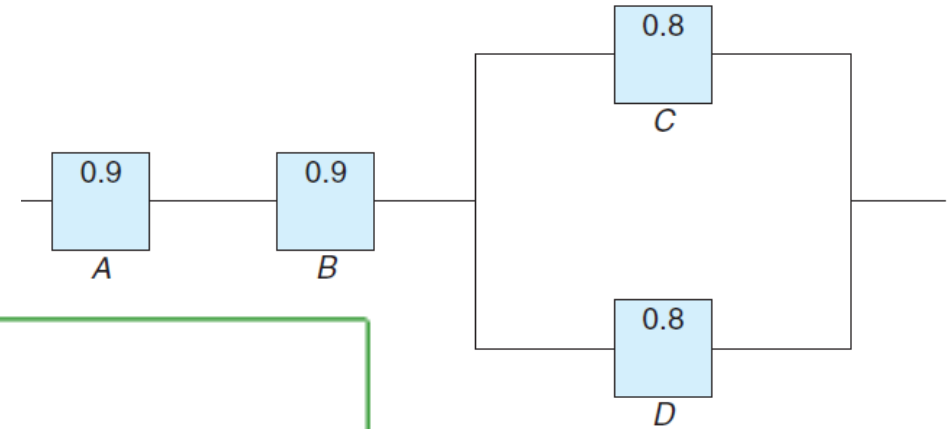


$$\begin{aligned} P &= \frac{P(\text{the system works but } C \text{ does not work})}{P(\text{the system works})} \\ &= \frac{P(A \cap B \cap C' \cap D)}{P(\text{the system works})} = \frac{(0.9)(0.9)(1 - 0.8)(0.8)}{0.7776} = 0.1667. \end{aligned}$$

Example: 4

An electrical system consist of four components. The system works if components **A and B work and either of the components C or D works**. The reliability of each component is also known. Assume that the four components work independently. Find the probability that

i. The entire system works.



$$P[A \cap B \cap (C \cup D)] = P(A)P(B)P(C \cup D)$$

$$= P(A)P(B)[1 - P(C' \cap D')]$$

$$= P(A)P(B)[1 - P(C')P(D')]$$

$$= (0.9)(0.9)[1 - (1 - 0.8)(1 - 0.8)]$$

$$= 0.7776$$

Conditional Probability

- ❖ Probability structure under some restrictions
- ❖ Probability of an event depend on the occurrence or non occurrence of another event

If in an experiment the events A and B can both occur, then

$$P(A \cap B) = P(A)P(B|A), \text{ provided } P(A) > 0.$$

The conditional probability of B , given A , denoted by $P(B|A)$, is defined by

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \quad \text{provided } P(A) > 0.$$

Example: 1

The probability that a regularly scheduled flight departs on time is $P(D) = 0.83$; the probability that it arrives on time is $P(A) = 0.82$; and the probability that it departs and arrives on time is $P(D \cap A) = 0.78$. Find the probability that a plane

- a) Arrives on time, given that it departed on time
- b) Departed on time, given that it has arrived on time

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$\begin{aligned} P(A|D) &= \frac{P(A \cap D)}{P(D)} \\ &= \frac{0.78}{0.83} \end{aligned}$$

$$\begin{aligned} P(D|A) &= \frac{P(A \cap D)}{P(A)} \\ &= \frac{0.78}{0.82} \end{aligned}$$

Example: 2

Ali tosses two fair dice. What is the conditional probability that

i. The sum of two dice will be 7 (D) given that the sum is odd (A)

$$S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), \\ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$$
$$D = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$$
$$A = \{(1,2), (1,4), (1,6), (2,1), (2,3), (2,5), (3,2), (3,4), (3,6), \\ (4,1), (4,3), (4,5), (5,2), (5,4), (5,6), (6,1), (6,3), (6,5)\}$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$A \cap D = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$$

$$P(D|A) = \frac{P(D \cap A)}{P(A)} = \frac{6/36}{18/36}$$

Example: 2

Ali tosses two fair dice. What is the conditional probability that the sum of two dice will be 7 (D) given that the sum is greater than 6 (B)

$$S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), \\ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$$
$$D = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$$
$$B = \{(1,6), (2,5), (2,6), (3,4), (3,5), (3,6), (4,3), (4,4), (4,5), (4,6), \\ (5,2), (5,3), (5,4), (5,5), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$B \cap D = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$$

$$P(D|B) = \frac{P(D \cap B)}{P(B)} = \frac{6/36}{21/36}$$

Example: 2

Ali tosses two fair dice. What is the conditional probability that the sum of two dice will be 7 (D) given that the two dice had the same outcome (C)

$$S = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), \\ (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), \\ (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), \\ (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), \\ (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), \\ (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$$

$$D = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$$

$$C = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$C \cap D = \{\}$$

$$P(D|C) = \frac{P(D \cap C)}{P(C)} = \frac{0/36}{6/36}$$

Example: 3

An urn contains two black balls and three white balls. Two balls are selected at random from the urn without replacement and sequence of colours is noted.

- i. Find the probability that both balls are black.

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A | B)P(B)$$

$$P(\text{both black balls}) = P(B_1 \text{ and } B_2)$$

$$= P(B_1)P(B_2|B_1)$$

$$= \left(\frac{2}{5}\right)\left(\frac{1}{4}\right)$$



Example: 3

An urn contains two black balls and three white balls. Two balls are selected at random from the urn without replacement and sequence of colours is noted.

- i. Find the probability that second ball is black.

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A | B)P(B)$$

$$P(\text{second black ball}) = P(B_1 \text{ and } B_2) \text{ or } P(W_1 \text{ and } B_2)$$

$$= P(B_1 \cap B_2) + P(W_1 \cap B_2)$$

$$= P(B_1)P(B_2|B_1) + P(W_1)P(B_2|W_1)$$

$$= \left(\frac{2}{5}\right)\left(\frac{1}{4}\right) + \left(\frac{3}{5}\right)\left(\frac{2}{4}\right)$$



Task - 1

The foundation of a tall building may fail because of inadequate bearing capacity or by excessive settlement. Let B and S represent the respective modes of foundation failure, and assume $p(B)=0.001$, $p(S)=0.008$ and $p(B|S)=0.1$, which is the conditional probability of bearing capacity failure given that there is excessive settlement.

1. Find the probability of failure of the foundation.
2. Find the probability that the building will experience settlement but without bearing capacity.

Task -2

There are two highways from City A to City B. Route 1 is on flat terrain, whereas Route 2 is a scenic route that goes through mountainous terrain. During severe winter seasons, one or both routes may be closed to traffic because of heavy snowfalls. Route 2 is obviously more likely to be closed than Route 1 during the winter months. The condition of Route 1 during a severe storm may depend on whether or not Route 2 is open to traffic.

Suppose, during a severe storm, the probabilities that the routes will be open are $p(R1)=0.75$, $p(R2)=0.50$, $p(R1R2)=0.40$

1. If Route 2 is open during a snow storm, find the prob. that Route 1 is also open.
2. If Route 2 is closed during a severe snow storm, find the probability that Route 1 is also closed.

